

Of course, it is not so easy to “falsify,” i.e., to state that something is wrong with full certainty. Imperfections in your testing method may yield a mistaken “no.” The doctor discovering cancer cells might have faulty equipment causing optical illusions; or he could be a bell-curve-using economist disguised as a doctor. An eyewitness to a crime might be drunk. *But it remains the case that you know what is wrong with a lot more confidence than you know what is right.* All pieces of information are not equal in importance.

Popper introduced the mechanism of conjectures and refutations, which works as follows: you formulate a (bold) conjecture and you start looking for the observation that would prove you wrong. This is the alternative to our search for confirmatory instances. If you think the task is easy, you will be disappointed—few humans have a natural ability to do this. I confess that I am not one of them; it does not come naturally to me.*

Counting to Three

Cognitive scientists have studied our natural tendency to look only for corroboration; they call this vulnerability to the corroboration error the *confirmation bias*. There are some experiments showing that people focus only on the books read in Umberto Eco’s library. You can test a given rule either directly, by looking at instances where it works, or indirectly, by focusing on where it does not work. As we saw earlier, disconfirming instances are far more powerful in establishing truth. Yet we tend to not be aware of this property.

The first experiment I know of concerning this phenomenon was done by the psychologist P. C. Wason. He presented subjects with the three-number sequence 2, 4, 6, and asked them to try to guess the rule generating it. Their method of guessing was to produce other three-number sequences, to which the experimenter would respond “yes” or “no” depending on whether the new sequences were consistent with the rule. Once confident with their answers, the subjects would formulate the rule. (Note the similarity of this experiment to the discussion in [Chapter 1](#) of the way history presents itself to us: assuming history is generated according to some logic, we see only the events, never the rules, but need to guess how it works.) The correct rule was “numbers in ascending order,” nothing more. Very few subjects discovered it because in order to do so they had to offer a series in descending order (that the experimenter would say “no” to). Wason noticed that the subjects had a rule in mind, but gave him examples aimed at confirming it instead of trying to supply series that were inconsistent with their hypothesis. Subjects tenaciously kept trying to confirm the rules that *they* had made up.

This experiment inspired a collection of similar tests, of which another